UNIT - I SUB - CODE - BOE-070

PROJECT MANAGEMENT

Topics :-

- Characteristics, types of projects
- PMBOK, role of Project manager, qualities
- Project organization and benefits
- Idea generation, needs of society,
- import substitution.
- Project lifecycle, project charter, project sponsor.
- Customer needs, stake holder concept,
- project scope, feasibility study and report
- baseline plan, SWOT analysis
- Project organization structure and hierarchy.
- Project teams, formation,
- attitude and aptitude

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⇒ What is a project ?

- A project is a temporary work taken up to create something new - like a product, service or result.

- It has a start date, and end date, specific goals, limited resources, and it is different from routine daily work.

⇒ Characteristics of a project

1. Temporary - Every project has a clear start and end date
2. Unique output - The result of a project is new
3. Specific goals - A project is done to achieve a well-defined objective.
4. Limited resources - Time, money, people and material are fixed
5. Planned work - Needs proper planning and execution
6. Uncertainty / risks - Projects always involve some challenges or risks
7. Teamwork - People with different skill often work together
8. Customer focus - The project is done to satisfy a client or end user.

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⇒ Types of Projects :-

1. Based on Purpose / output
   - Construction Project - e.g building bridges, houses, roads
   - IT / Software Project - e.g developing apps, website
   - Research Project - e.g finding a new medicine, make research
   - Business Project - e.g launching a new product, marketing campaigns.
   - Social Projects - eg NGO programs, awareness campaigns.

2. Based on Size
   - Small project - Done with few people and limited time
   - Medium project - Require more planning and resources
   - Large project - Complex, long-term, many teams involved

3. Based on Time duration
   - Short-term projects - Finish in weeks or months
   - long-term projects - Take years to complete

4. Based on Complexity
   - Simple projects - Easy tasks, fewer risks
   - Complex projects - Many activities, high risk, advanced planning needed

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⇒ Project Management Body of Knowledge (PMBOK)

- PMBOK = Project Management Body of Knowledge
- It is a guidebook created by PMI that explains how to manage projects properly
- It is like a recipe book for project managers

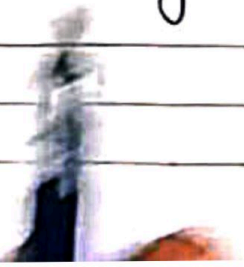
1. **Standard Guidelines for Projects**
PMBOK gives a set of rules, methods, and best practices to run projects successfully.

2. **Process - Based Approach**
It divides project Management into processes instead of random tasks.

3. **Five Process Groups** (Main stages of a Project)

1. Initiating :- start the project, get approvals
2. Planning :- Make a roadmap (budget, timeline, risk, tasks).
3. Executing :- Do the actual work
4. Monitoring & Controlling :- Track progress, fix issues
5. Closing :- Finish the project, hand over to client

4. **Ten Knowledge Areas**
PMBOK explains 10 areas you must manage in every project:



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1. Integration - Making sure all parts of the project work together.
2. Scope - Deciding what is included / excluded.
3. Time - Managing deadlines.
4. Cost - Managing budget.
5. Quality - Delivering a good product
6. Human Resources - Managing the team
7. Communication - sharing info clearly.
8. Risk - Handling uncertainties
9. Procurement - Buying external stuff.
10. stakeholders - Managing people affected by the project

5. Focus on Best Practices
PMBoK doesn't say "do it my way", it gives best practices so you can adjust based on your project

⇒ Project manager

- A project manager is the person responsible for leading a project from start to finish.
- A project manager's main role is to plan, execute, and finish projects on time and within budget. They are the person who makes sure everything gets done correctly

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Key Roles of a Project Manager

- **Planning** :- They create a detailed plan, including project goals, timelines, and budgets
- **Leading a Team** :- They organize, motivate, and manage the team to get the work done.
- **Managing Resources** :- They make sure the team has the right people, tools and money
- **Handling Problems** :- They find and solve any issues or risks that come up during the project
- **Communicating** :- They keep everyone, from the team to the client, informed about the project's progress

Key Qualities of a Project Manager

- **Leadership** : They inspire and guide the team
- **Communication** : They can clearly explain things to differents people
- **Problem - Solving** : They can quickly find solutions to unexpected issues.
- **Organization** :- They are great at keeping track of tasks, deadlines and details
- **Adaptability** : They can handle changes and unexpected twists without losing control

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=> **Project Organization :-**

Project organization means how a project is structured and managed - who will do what, how tasks will be divided and how communication will flow

Types of Project Organization

1. **Functional Organization :-** Project work is done with existing departments (like, IT, HR, Marketing)
2. **Projectized Organization :-** A separate team is created only for the project, with a project manager in full control
3. **Matrix Organization :-** Combination of functional + projectized (team member reports to both their department head and project manager)

=> **Benefits of Project Organization**

- **Clear Focus :** Everyone on the team is dedicated to one single goal. This prevents distractions and ensures all efforts are aligned
- **Faster Decisions :-** The project manager has direct authority over the team, which allows for quicker - decision making without waiting for approvals from multiple department

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- **Better Communication :-** Since the team members are all working together, communication is more direct and effective.
- **Efficiency :-** Better ways of doing work
- **Economic growth :-** creates jobs, increases business opportunities

⇒ **Idea Generation :-** Idea generation is the creative process of producing and developing new ideas or concepts

⇒ **Brainstorming :-**
- People sit together and freely share their ideas without criticism
- The focus is on quantity of ideas first, then quality

⇒ **Market Research :-**
- Collecting data about what customers want like, dislike or lack in current products
- Can be done → surveys, interviews, competitor analysis or social media trends

⇒ **Observation - Looking at problems in daily life**
- Simply watching real-life situations and noticing where people face problems
- Many successful projects come from day-to-day problem solving

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4. Technology Trends - Using New Tech for New sol^n.
- Watching emerging technologies and applying them to projects
- Helps in creating modern, innovative solutions that people will adopt

5. Customer Feedback - Suggestions from users
- Listening to what user say about your product / service.
- Helps improve existing products or create new ones

⇒ Import Substitution:-
- A strategy where a country tries to replace imported goods with locally made goods
- Its main goal is to reduce dependence on foreign countries, promote self-reliance and save foreign currency
- Common in developing countries to strengthen domestic industries

⇒ Project life cycle

The project life cycle is a series of phases a project goes through from its begining to its end.

1. Initiation
- This is the starting point.
- The main goal here is to determine if the project is feasible and worth doing

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2. Planning
This is where you build the project roadmap.

- Scope :- what work will be done and what won't
- Schedule :- A timeline with all tasks and deadlines
- Budget :- An estimate of the costs
- Resources :- who will do the work and what tools they'll need.
- Risks :- what could go wrong and how to handle it.

3. Execution
- This is the "doing" phase.
- The team starts working on the task outlined in the path.
- The project manager's job here is to manage the team, allocate resources, and keep everything moving forward

4. Monitoring & Controlling
- This phase runs alongside the execution phase.
- The project manager constantly tracks the project's progress to make sure it's on schedule and within budget.

5. Closure :-
Once all the project's work is complete,

this phase officially closes it.
→ Delivering the final product.
→ Getting final approval.

⇒ **Project charter :-**

- A formal written document that gives permissi to start the project
- prepared at the Initiation stage.
- Authorizes the project Manager to use resources

⇒ **It contains :-**

1. Project name & description
2. Purpose / why (business need)
3. Scope (what is included / excluded)
4. Timeline & budget
5. Key Stakeholders (sponsor, client, manager)
6. Risks & assumptions
7. Authority of project manager.

⇒ **Project Sponsor :-**

- The main supporter & fund provider of a project
- usually a senial leader, client, or organization

Role :-
→ Provide money & Resources
→ Approves project charter & Scope.
→ Supports the project manager when issues arise
→ Ensure project aligns with business goals

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⇒ Customer Needs :-

- Customer needs are the requirements, problem or expectations that customer want to be fulfilled by a product or service.
- If a company understands customer needs well → it can design better products, give better service and stay ahead of competitors

Types of customer Needs

1. Basic Needs
→ The minimum expectation
→ If not fulfilled → customer will be dissatisfied

2. Performance Needs
→ Customers want better performance, speed, reliability

3 Delight Needs
→ Things customers don't expect, but if given, they feel very happy

⇒ Stakeholder

- A stakeholder is any person, group or organization that has interest in a project and is affected by its outcome.
- They can influence the project or be influenced by it.

⇒ Types of stakeholders

1. External stakeholders (outside the organization)
   - Customers / End users
   - Government
   - Suppliers
   - Community / society

2. Internal stakeholders (inside the organization)
   - Employees
   - Managers
   - Owners / shareholders

⇒ Project Scope :-

- Project scope means what the project will include and what it will not include
- It clearly defines the boundaries of the project - what work needs to be done, what the final result will be and what is outside the project.

Key Element : -

1. Objectives - what the project wants to achieve.
2. Deliverables - Final outputs
3. Tasks / activities - Work to be done
4. Boundaries - what is not part of the project
5. Constraints - limits of time, cost, resources
6. Acceptance criteria - Conditions for projects to be considered "successfully done".



⇒ **Feasibility study :-**

- Feasibility study is an analysis done before starting a project to check whether the project is possible, practical and worth doing.
- It answers the question : "should we go ahead with this project or not ?"

Types of Feasibility :-

1. **Technical Feasibility :-**
   Do we have the technology, tools and skills ?
2. **Economic feasibility :-**
   Will the project be profitable
3. **Legal feasibility :-**
   Is the project allowed by law and regulations ?
4. **Operational Feasibility :-**
   Will the project actually work in real life ?
5. **Schedule Feasibility :-**
   Can we finish the project within the deadline

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⇒ **Report :-**

A report is a formal document that present information, findings, analysis, or progress about a project

- It is usually written in a structured way for decision - making.
- It helps communicate project status to stakeholders

Format :-

1. Title Page - Project name, date, prepared by.
2. Executive Summary - Quick overview of the report
3. Introduction - Purpose of the project / report
4. Objectives - what the project aims to achieve
5. Methodology - How the work was done.
6. Finding / Status - Results, progress, or current situation
7. Problems / challenges - Issue faced.
8. Recommendations - Suggestions for improvement
9. Conclusion - Final Summary
10. Appendices - Extra details (data, charts, graphs)

⇒ **Baseline Plan :-**

- A baseline plan is the original approved plan for a project.
- It is like a reference point or fixed snapshot of the project's scope, time, cost and resources.
- later, actual progress is compared against this baseline to check whether the project is on track / delayed, over budget

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⇒ SWOT Analysis :-

SWOT stands for :
- S → Strengths
- W → Weakness
- O → Opportunities
- T → Threats

- It is a strategic planning tool used to understand both internal and external factors that affect a business, project, or, idea.
- It helps in making better decisions and strategies.

Elements of SWOT :-

1. **Strengths**
   - → Things the organization does well.
   - → Resources, skills or advantages
2. **Weakness**
   - → Areas where improvement is needed.
   - → Limitations, lack of resources, or poor resources
3. **Opportunities**
   - → External chances to grow, expand or gain advantage.
   - → Market gaps, trends, or new demands
4. **Threats**
   - → External challenges or risks that could harm the business

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. Competition, market changes, or government rules

⇒ Project Organization Structure

. A project organization structure shows how peopl roles and responsibilities are arranged in a project.
. It define who report to whom and who is responsible for what.
. It helps in clear communication, decision-making and smooth workflow.

⇒ Types of Project Organization Structure

1. Functional structure :-
. Team members are grouped by specialization / departm (like IT, HR, Finance)
. The functional manager has authority
. Projects get resources from these departments

2. Projectized Structure :-
. Entire team works only for the project
. Project manager has full authority (budget, resources, decisions).
. Teams report directly to the project manager

3. Matrix structure (Mix of both)
. Combines functional + Projectized
. Team members report to both functional manager and project manager.

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⇒ **Project Hierarchy :-**

⇒ Hierarchy means the levels of authority and responsibility in a project.
→ It shows who make decisions, who manages and who does the actual work.

1. **Top level** - Project Sponsor / client
   - Provide funding, resources, and approval
   - Defines overall goal of the project.
   - Has the highest authority.

2. **Upper Management** - steering committee
   - A group of senior stakeholders (director, senior manager)
   - Helps the sponsor in making key decisions
   - Reviews progress at important stages

3. **Middle level** - Project Manager
   - The captain of the project
   - Responsible for planning, scheduling, controlling and delivering.
   - Manages team, monitors risks, reports to sponsor

4. **Lower - Middle level** - Team leaders / Functional Managers
   - Handle specific sections of the project
   - Guide small teams (developers, tester, designers)
   - Report to Project Manager.

5. **Operational level** - Project team Members
   - Do the actual work of the project

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- Follow instructions from team leads or project manager

6. Support Roles
- Provide indirect help.
- Includes HR, Finance (budget control), Procuremen (buying materials)

⇒ Project Teams

- A project team is a group of ~~peo~~ people brough together to complete a project.
- Each member has specific roles (manager, develope tester, designer, etc).
- It can include Internal employees, external consul tants, vendors and stakeholders

⇒ Team formation

1. Forming :-
- Team members come together.
- Roles, responsibilities, and goals are introduced

2. Storming :-
- Members start working → conflicts may arise
- Agruments about methods, tools or deadlines

3 Norming :-
- Team agrees on rules, build trust.
- Everyone starts working in harmony

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4. Performing
- Teams works efficiently, minimum conflict.
- High productivity stage.

5. Adjourning
- Project ends → team is dissolved.

⇒ Attitude in Project Teams

- The mindset, behavior, and approach team members bring to work.
- It affects how people react to problems, interact with others, and handle stress.

Types :-

1. Positive Attitude
- Focused on solutions, cooperation and responsibility
- Build trust, motivation, and team spirit.
- Helps the team overcome challenges faster.

2. Negative Attitude
- Leads to blame, resistance, and lack of commitment.
- creates conflicts, stress, and project delays

⇒ Aptitude in Project Teams

- The skills, abilities, knowledge, and talent of team members
- It decides what a person can do and how well they can perform tasks

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Types of Aptitude in Project Teams

1. **Technical Aptitude**
   - Core professional abilities
   - Example : Developer, tester, Designer, Engineer

2. **Soft Skills Aptitude**
   - Skills for communication, teamwork and leadership
   - Example : Project Manager, Team Members etc

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